

CARLOFORTE, Italy

WMO: 165490

Lat: 39.137N

Lon: 8.312E

Elev: 12

StdP: 101.18

Time Zone: 1.00 (EUC)

Period: 86-08

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	7.0	2.8	4.6	19.4	3.6	4.9	21.7	13.8	13.4	12.4	12.6	3.4	70	0.417

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.4	31.9	18.1	30.8	18.2	29.7	18.3	24.6	27.6	23.7	26.4	22.8	25.5	3.6	320

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	18.6	26.1	22.8	17.5	25.6	21.8	16.5	24.8	74.7	27.4	70.9	26.6	67.4	25.7	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.4	9.1	4.2	35.1	1.5	1.1	3.1	35.8	2.2	36.5	1.4	37.1	0.3	37.8		
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	11.9	12.5	13.5	15.0	19.0	22.4	25.4	25.9	23.1	20.2	15.9	12.9
	DBStd	5.40	1.97	1.82	2.02	2.00	2.37	2.37	1.91	1.66	2.36	2.61	2.23	2.17
	HDD10.0	17	8	3	2	0	0	0	0	0	0	0	0	4
	HDD18.3	892	200	164	149	100	20	2	0	0	1	12	77	168
	CDD10.0	3001	66	72	111	151	279	372	476	492	393	318	176	94
	CDD18.3	835	0	0	0	1	41	123	218	233	144	71	3	0
	CDH23.3	5258	0	0	0	4	133	686	1694	1818	733	189	1	0
	CDH26.7	1433	0	0	0	1	9	134	549	591	139	11	0	0
Wind	WSAvg	3.7	3.6	3.5	3.8	4.3	3.4	3.3	3.4	3.5	3.4	3.6	4.1	4.4
Precipitation	PrecAvg	569	59	57	47	55	33	16	6	10	36	70	97	83
	PrecMax	805	152	119	99	125	75	67	15	43	112	167	237	156
	PrecMin	380	17	11	8	13	4	2	2	1	3	15	23	26
	PrecStd	107	32	31	24	29	19	16	4	9	26	36	49	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.6	18.8	20.8	22.6	27.9	31.1	34.0	33.5	31.0	28.0	22.8	19.6
		MCWB	14.6	12.5	13.3	13.4	15.8	16.6	17.2	17.5	19.3	16.7	18.2	16.8
	2%	DB	16.4	17.5	19.2	21.0	26.0	29.6	32.1	32.1	29.4	26.4	21.4	18.1
		MCWB	13.4	12.6	12.8	14.0	15.2	17.1	18.7	17.6	18.5	17.5	16.2	14.1
	5%	DB	15.6	16.6	18.2	19.9	24.9	28.2	30.8	30.9	28.3	25.4	20.4	17.0
		MCWB	12.9	12.5	12.5	13.7	15.4	16.6	18.9	17.7	18.5	17.9	15.3	13.9
	10%	DB	14.9	15.7	17.1	18.8	23.5	27.0	29.7	29.8	27.2	24.3	19.3	16.0
		MCWB	12.5	12.2	12.3	13.4	14.8	15.8	19.2	18.0	18.1	18.0	14.8	12.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.3	15.8	16.2	17.4	20.4	23.7	25.9	25.1	24.8	22.5	19.8	17.5
		MCDB	17.5	16.7	18.1	19.5	23.6	27.1	28.5	28.6	27.0	24.4	21.8	18.4
	2%	WB	15.0	14.7	15.1	16.6	18.9	22.6	25.0	24.1	23.7	21.6	18.4	16.2
		MCDB	15.9	15.8	16.9	18.7	22.3	25.0	28.2	26.9	26.1	23.5	20.4	17.0
	5%	WB	14.3	13.9	14.1	15.9	18.1	20.0	24.0	23.2	22.5	20.7	17.3	15.1
		MCDB	15.2	15.0	15.2	17.9	21.1	24.5	26.8	26.0	25.2	22.5	19.1	16.0
	10%	WB	13.5	12.9	13.1	15.0	17.2	17.2	23.0	22.3	21.2	19.7	16.1	13.9
		MCDB	14.4	14.0	14.7	17.1	20.3	24.1	25.8	24.9	23.6	21.4	17.7	15.0
Mean Daily Temperature Range	5% DB	MDBR	5.6	6.5	6.6	6.9	8.0	8.4	8.8	8.4	7.4	6.3	5.7	5.4
		MCDBR	6.4	7.4	8.0	8.5	9.5	9.3	10.2	9.6	8.0	7.1	7.0	6.0
		MCWBR	3.8	3.5	3.1	3.2	3.0	2.9	3.6	3.3	3.0	3.2	3.4	3.3
	5% WB	MCDBR	5.8	6.2	6.5	6.8	7.4	7.6	8.2	8.4	7.5	6.5	6.7	5.7
		MCWBR	4.8	4.7	4.4	4.2	4.2	5.2	4.9	5.1	4.5	4.0	4.5	4.7
Clear-Sky Solar Irradiance	taub	0.344	0.364	0.402	0.438	0.453	0.432	0.448	0.437	0.433	0.409	0.380	0.353	
	taud	2.468	2.400	2.290	2.205	2.199	2.306	2.265	2.305	2.307	2.345	2.375	2.432	
	Ebn at Noon	822	852	851	841	833	849	832	832	811	789	766	779	
	Edh at Noon	81	98	121	140	144	129	134	125	118	103	88	79	
All-Sky Solar Radiation	RadAvg	2.09	2.99	4.38	5.59	6.65	7.65	7.67	6.76	5.10	3.66	2.32	1.88	
	RadStd	0.17	0.28	0.40	0.36	0.42	0.24	0.24	0.29	0.19	0.25	0.16	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (7 neighbors)	+0.28	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page